

DONGU KIM

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TECHNICAL INTERESTS

Distributed storage system, LLM serving system, Serverless, Kernel-bypass, P4 switch, Accelerator

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) 2022 - 2024
M.S in Computer Science and Engineering

Inha University 2015 - 2022
B.S. in Electronic Engineering

EXPERIENCES

Mangoboost Inc. April 2024 - Present
System Software Engineer

- **Software Defined Storage (SDS): Scalable, High-Performance System for AI Clusters**
 - Designing and developing a storage-agnostic backend system to automate volume provisioning, eliminating manual cluster-level interventions.
 - Optimizing storage systems for KV Cache to accelerate Large Language Model (LLM) serving performance and reduce inference latency.
- **NVMe-over-TCP Initiator (NTI): Hardware-offloaded Virtualized NVMe Accelerator**
 - Achieved UNH-IOL NVMe Certification, ensuring full interoperability and OS-agnostic control path operations across Windows, Linux, and VMware.
 - Developed robust failover and resilience mechanisms for high availability during network disruptions, while abstracting the TCP stack for seamless user integration.

Next Generation Networks and Systems Lab (Advisor: Youngbin Im) Mar 2022 - Feb 2024
Research Assistant

- **FlexServer: High-performance Content Servers Accelerated by DPU (400 Gbps Throughput)**
 - Masked I/O latencies inherent in the legacy Linux stack by implementing a fully asynchronous architecture leveraging `io_uring`.
 - Accelerated data access by prefetching high-popularity content into DPU memory, effectively exploiting Zipf-like content request patterns.

Intelligent Cloud & Network Lab (Advisor: YeongJin Kim) Mar 2021 - July 2021
Undergraduate Research Intern

- **Core Principles of Cloud Computing**
 - Studied cloud infrastructure fundamentals, focusing on resource virtualization, SDN (Software Defined Networking), and storage orchestration.

PATENTS

1. Youngbin Im and Dong U Kim, "METHOD AND APPARATUS FOR IMPROVING KEY VALUE STORE PERFORMANCE THROUGH OFFLOADING OF SMART NETWORK INTERFACE CARD," KR Patent Registered No. 10-2724-6120000, Oct. 2024

TECHNICAL SKILLS

- **Languages:** C/C++, Go, Python, P4
- **Technologies:** DAOS, Ceph, DPDK, SPDK, DOCA, NVMeoF, RDMA, TCP, Kubernetes, Pytorch
- **Hardware Accelerator:** FPGA, SmartNIC/DPU (Bluefield, Agilio SmartNIC)
- **CICD:** Docker, Debian, RPM, Git, Github-Action

EXTRACURRICULAR ACTIVITIES

Competitions & Projects

- **The World Embedded Software Contest 2019** *April 2019 - Dec 2019*
 - Developed multithreaded software for an autonomous RC car featuring real-time obstacle avoidance; gained in-depth expertise in concurrent programming and thread synchronization
 - Integrated various sensors—including LiDAR and Ultrasonic—with the software architecture to improve the accuracy of environmental mapping

Volunteer works

- **Freview** *Feb 2020 - Jan 2021*
 - Served as a volunteer mentor, sharing specialized knowledge about my major to help middle and high school students explore potential career paths and gain a deeper understanding of the field